INTELLIGENT PERSONAL DATA LEAKAGE DETECTION SYSTEM USING DEEP LEARNING AND RULE-BASED ANALYSIS

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TCR25MCA-2003

Mini Project

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Outline

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Introduction

- Rapid growth of digital communication platforms (email, cloud storage, messaging, enterprise systems) has increased the risk of personal data leakage.
- Sensitive information such as passwords, financial data, PII, and confidential organizational data is often unintentionally or maliciously exposed.
- Rule-based and signature-based systems detect only predefined patterns and lack context awareness.
- Conventional machine learning models lack deep contextual understanding of unstructured text.
- Encryption and access control mechanisms fail to prevent insider threats once legitimate access has been granted.

Objectives:

To design an Intelligent Personal Data Leakage Detection System that combines rule-based filtering, transformer-based NLP contextual analysis, deep learning (LSTM/CNN) pattern detection, and anomaly-based insider-threat detection within a single hybrid pipeline, accompanied by an explainable AI layer so that every detection decision can be understood and acted upon by a security analyst.

Guan, Z., and Chen, J., “Research on Data Leakage Detection and Protection Based on Deep Learning,” *Proc. 2024 IEEE Int. Conf. on Big Data & Machine Learning (ICBDML)*, 2024.

- Applied deep learning techniques to detect and protect against sensitive data leakage in cloud environments.
- Quantitatively evaluated leakage risk and reported accuracy and recall of the detector.
- Used learned representations to identify sensitive information even without an exact pattern match.
- **Observation:** Content-focused; does not reason about user behaviour or insider-threat indicators.

Alzaabi, F. R., and Mehmood, A., “A Review of Recent Advances, Challenges, and Opportunities in Malicious Insider Threat Detection Using Machine Learning Methods,” *IEEE Access*, vol. 12, pp. 30907–30927, 2024.

- Comprehensive review of machine-learning-based malicious insider-threat detection.
- Consolidated recent advances, open challenges, and opportunities across the field.
- Surveyed techniques for modelling and flagging anomalous user behaviour.
- **Observation:** Behavioural focus only; not linked to content-based leakage signals.

Literature Review

Mainetti, L., and Elia, A., “Detecting Personally Identifiable Information Through Natural Language Processing: A Step Forward,” *Applied System Innovation*, vol. 8, no. 2, p. 55, 2025.

- Proposed a BERT-based NLP approach to personal data leakage detection.
- Identified sensitive personal information based on semantic content rather than surface form alone.
- Directly relevant to contextual, semantic-level analysis of unstructured text.
- **Observation:** Content-level only; does not integrate rule-based filtering or anomaly-based signals.

Literature Review

Santhosh, S., Revanth, M., Seshatri, M., Jayavelu, K., Sujitha, M., and Nisha, M., “Enhanced Data Leakage Detection in Cloud and Enterprise Environments through NLP-Driven Classification, Metadata Tagging, and Anomaly-Based Learning,” *Int. J. Res. Appl. Sci. Eng. Technol. (IJRASET)*, 2025.

- Proposed an enhanced, hybrid real-time DLP approach for cloud and enterprise environments.
- Combined NLP-driven content classification, metadata tagging, and anomaly-based learning.
- Improved detection responsiveness and coverage over single-technique approaches.
- **Observation:** No explicit rule-based filtering stage or dedicated explainability layer.

Literature Review — Comparative Analysis

Author	Approach	Brief Observation
Guan and Chen (2024)	Deep-learning content inspection	Detects sensitive content without an exact pattern match; content-focused, no behavioural context.
Alzaabi and Mehmood (2024)	ML-based insider-threat review	Consolidates insider-threat ML techniques; behavioural focus only.
Mainetti and Elia (2025)	BERT-based NLP detection	Detects PII from unstructured text via context; content-level only.
Santhosh et al. (2025)	Hybrid real-time DLP	Combines textual and behavioural signals; no explicit rule-based stage.

Existing Systems

- Rule-based systems detect predefined patterns (email addresses, phone numbers) but lack semantic understanding of surrounding context.
- Signature-based detection is effective against known threat indicators but ineffective against new or evolving leakage patterns.
- Traditional machine learning approaches struggle with unstructured, highly contextual data.
- Advanced deep learning or NLP-based detectors are typically applied in isolation, without rule-based filtering or behavioural anomaly detection in the same pipeline.

- An Intelligent Personal Data Leakage Detection System combining rule-based filtering, transformer-based NLP contextual analysis, deep learning (LSTM/CNN) pattern detection, and anomaly-based insider-threat detection within a single hybrid pipeline.
- Accepts text data and documents as input, applies each detection component in turn, and combines their outputs into a unified risk score and classification.
- An explainable AI layer accompanies every detection decision with a human-readable rationale.
- A real-time monitoring component generates alerts as sensitive data leakage is detected, viewable through a monitoring dashboard.

Hardware Requirements

- **Processor**: Intel Core i5 / AMD Ryzen 5 or higher
- **Memory**: Minimum 8 GB RAM (16 GB recommended)
- **GPU (optional)**: CUDA-capable GPU with at least 4 GB VRAM, to accelerate BERT and LSTM/CNN training
- **Storage**: 20–50 GB free disk space

Software Development Requirements

- **Operating System**: Windows 10 / Ubuntu 20.04 LTS / macOS
- **Back End**: Python 3.10 or higher
- **IDE Used**: Visual Studio Code
- **Version Control**: Git
- **Development Tools and Frameworks**:
  - PyTorch, Hugging Face Transformers (BERT)
  - spaCy, scikit-learn
  - Flask, SHAP / LIME

Functional Requirements

- **Data Input Handling:** The system shall accept text data and documents, including PDF files, and validate submitted input.
- **Rule-Based Filtering:** The system shall detect known, structured patterns such as email addresses, phone numbers, and financial-data patterns using configurable regex rules.
- **NLP-Based Contextual Analysis:** The system shall apply a transformer-based NLP model (e.g., BERT) to distinguish genuine sensitive-data leakage from superficially similar content.
- **Deep Learning-Based Pattern Detection:** The system shall apply LSTM/CNN-based models to detect complex or hidden leakage patterns.

- **Anomaly Detection for Insider Threats:** The system shall monitor data-handling behaviour and flag anomalies as potential insider-threat incidents.
- **Risk Scoring:** The system shall compute a risk score based on the type, sensitivity, and confidence of detections and classify each input accordingly.
- **Real-Time Monitoring and Alerts:** The system shall support near real-time processing and generate alerts when leakage is detected.
- **Explainable AI Reporting:** The system shall generate a human-readable explanation report for every detection decision.

Non-Functional Requirements

- **Accuracy:** The system must achieve high precision and recall while minimising false positives.
- **Performance:** The system must operate with low latency and high throughput for near real-time analysis.
- **Scalability:** The system must handle large-scale enterprise data volumes without disproportionate latency increase.
- **Security:** The system must ensure sensitive data processed for analysis is not itself exposed as a result of its own operation.
- **Reliability:** The system must handle malformed or unsupported input gracefully, without failing outright.
- **Maintainability:** Rule-based, NLP/deep-learning, anomaly-detection, and reporting components must remain independently maintainable.

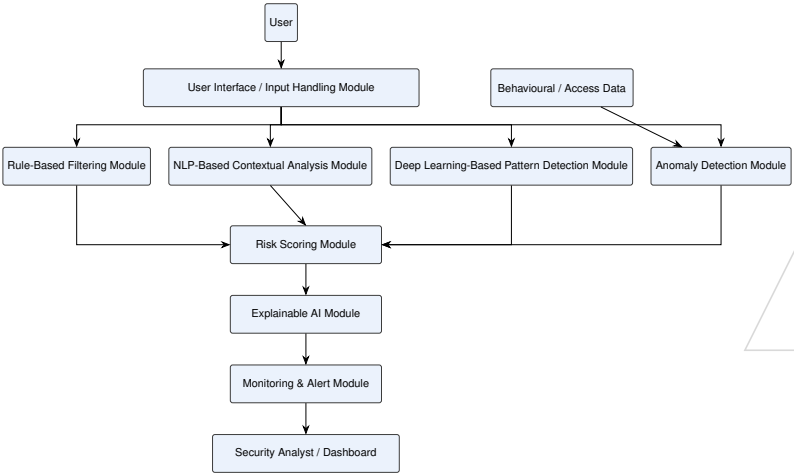
Input

- Text data and documents (including PDF files) submitted for analysis, drawn from emails, cloud storage, messaging applications, and enterprise systems.
- **Input Format:** Free-form and structured text content, along with document uploads validated by the system before processing.

Output

- A computed risk score and classification for each analysed input, real-time alerts and notifications for detected leakage, and an explanation report describing the reasoning behind each decision.
- **Output Format:** Alerts, risk scores, detected sensitive data, and explanation reports presented together through an interactive monitoring dashboard.

System Architecture



System Architecture

- The User submits text or documents, which pass through the User Interface Module for validation.
- Validated input is analysed in parallel by the Rule-Based Filtering, NLP-Based Contextual Analysis and Deep Learning-Based Pattern Detection modules.
- The Anomaly Detection Module separately monitors behavioural data for insider-threat indicators.
- All outputs converge at the Risk Scoring Module, followed by the Explainable AI Module and the Monitoring & Alert Module, before reaching the Security Analyst via the dashboard.

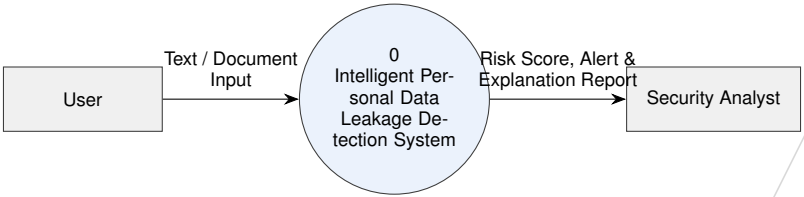
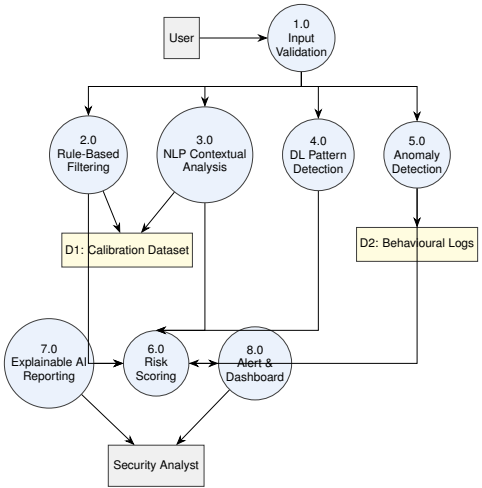
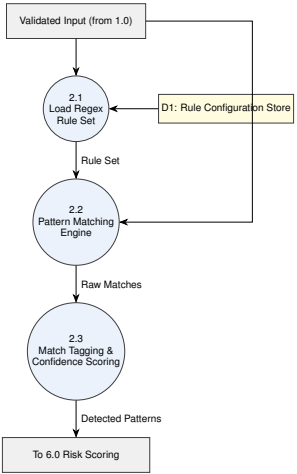


Figure 2: DFD Level 0

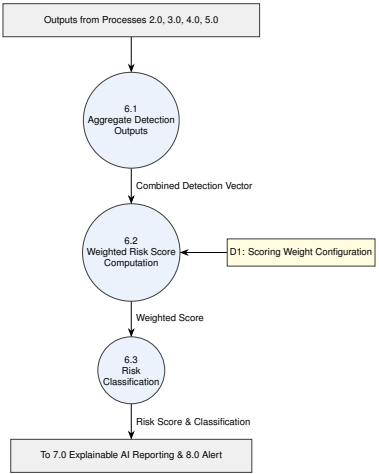
Data Flow Diagrams



Data Flow Diagram: Level 2 (Process 2.0)



Data Flow Diagram: Level 2 (Process 6.0)



System Modules

- **User Interface / Input Handling Module:** Accepts text data and documents, including PDFs, and validates submitted input before further processing.
- **Rule-Based Filtering Module:** Applies configurable regular-expression patterns to detect known, structured identifiers such as email addresses, phone numbers and financial data.
- **NLP-Based Contextual Analysis Module:** Uses a fine-tuned BERT model to analyse the semantic context of input, distinguishing genuine sensitive content from harmless, superficially similar text.
- **Deep Learning-Based Pattern Detection Module:** Applies LSTM/CNN models to detect complex or previously unseen leakage patterns not captured by rule-based or contextual analysis alone.

System Modules

- **Anomaly Detection Module:** Monitors data-handling behaviour for deviations from normal patterns, flagging potential insider-threat activity.
- **Risk Scoring Module:** Combines outputs from all detection components into a single, unified risk score and classification.
- **Explainable AI Module:** Uses SHAP and LIME to generate a human-readable rationale for every detection decision.
- **Monitoring and Alert Module:** Generates real-time alerts and presents risk scores, explanations and alerts to the security analyst via a Flask-based dashboard.

Conclusion

- This project proposes an Intelligent Personal Data Leakage Detection System that unifies rule-based filtering, transformer-based NLP contextual analysis, deep learning pattern detection, and behavioural anomaly detection within a single hybrid pipeline.
- Combining these complementary techniques allows the system to catch both structured, known leakage patterns and unstructured, previously unseen ones – while an explainable AI layer keeps every decision transparent and actionable for a security analyst.
- Overall, the system demonstrates how content-level and behaviour-level signals can be brought together in one pipeline to detect personal data leakage more comprehensively than any single technique used in isolation.

References

1. Guan, Z., and Chen, J., “Research on Data Leakage Detection and Protection Based on Deep Learning,” *Proc. 2024 IEEE Int. Conf. on Big Data & Machine Learning (ICBDML)*, 2024.
2. Alzaabi, F. R., and Mehmood, A., “A Review of Recent Advances, Challenges, and Opportunities in Malicious Insider Threat Detection Using Machine Learning Methods,” *IEEE Access*, vol. 12, pp. 30907–30927, 2024.

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3. Mainetti, L., and Elia, A., “Detecting Personally Identifiable Information Through Natural Language Processing: A Step Forward,” *Applied System Innovation*, vol. 8, no. 2, p. 55, 2025.
4. Santhosh, S., Revanth, M., Seshatri, M., Jayavelu, K., Sujitha, M., and Nisha, M., “Enhanced Data Leakage Detection in Cloud and Enterprise Environments through NLP-Driven Classification, Metadata Tagging, and Anomaly-Based Learning,” *Int. J. Res. Appl. Sci. Eng. Technol. (IJRASET)*, 2025.

Thank You